

Accounting and Budgetary Systems for Engineers - INFO7225

MGEN

Chapter 5. Process Costing System

Professor Shiaoming Shi
College of Engineering
Northeastern University





Process Costing

A **job costing information system** keeps track of the specific costs for raw materials, labor, and overhead associated with each product or group of products, called a “job.” This type of costing system is most appropriate for manufacturers of large-scale or custom products, such as home builders or book publishers. Manufacturers of homogeneous products (such as soft drinks or toothbrushes) that are produced on a regular and continuous basis use a **process costing information system**. In this system, it is not feasible or practical to keep track of costs for each item or group of items produced. Instead, process costing systems use averages to calculate the costs associated with goods in process and finished goods produced.

Activity-based costing systems help managers describe processes, identify cost drivers of each process, and then determine the unit costs of products created in each process. By studying their business processes, managers are in a better position to recognize opportunities to improve those processes. Thus, activity-based costing gives managers a better understanding of their processes, an improved ability to allocate indirect costs to those processes, and a better understanding of the true cost of each product. The systems flowchart in **Figure 9-7** shows a



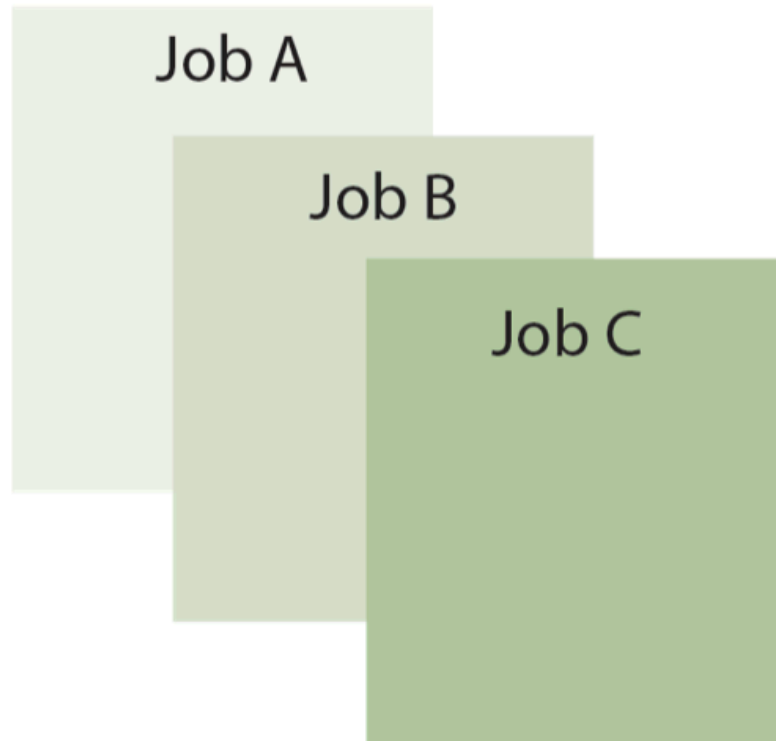
Job Order Costing vs. Process Costing

Job Order Costing versus Process Costing

	Job Order Cost System	Process Cost System
Product type	Custom order	Mass production
Examples	Signs, buildings, tax returns	Folding tables, toys, buffet restaurants
Cost accumulation	Job lot	Accumulated per process
Work in process inventory	Individual job cost sheets	Separate work in process inventory department
Record keeping	Individual job cost sheets	Production cost report

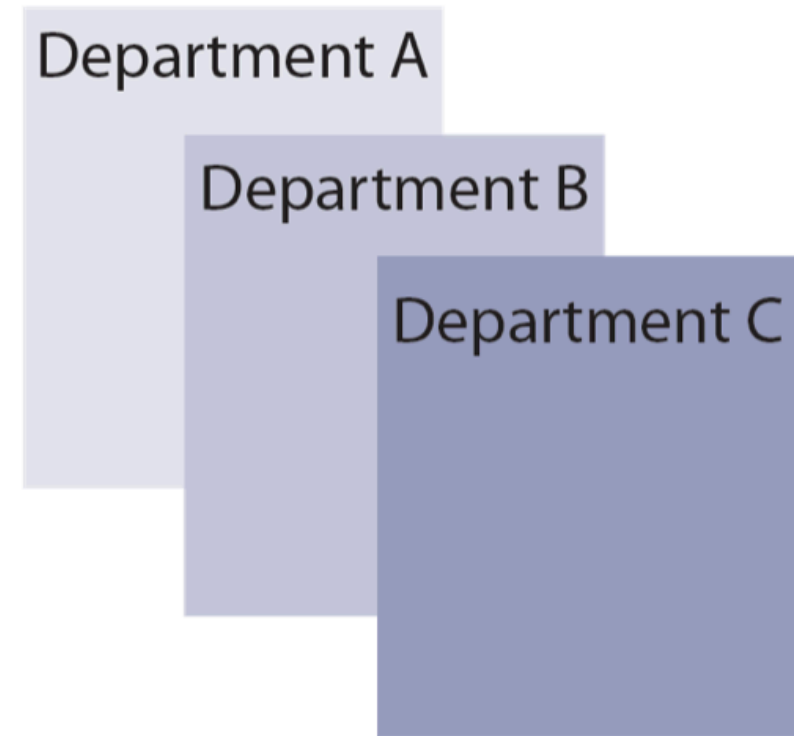
Job Costing Environment

Costs are accumulated
by job on Job Cost Sheets



Process Costing Environment

Costs are accumulated by department
on Cost of Production Reports



Job Costing Flows

Raw
Materials

|

Work in Process
Job A

|

Finished Goods
Job A

|

Wages
Payable

|

Work in Process
Job B

|

Finished Goods
Job B

|

Factory
Overhead

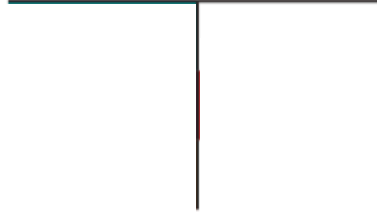
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Process Costing Flows

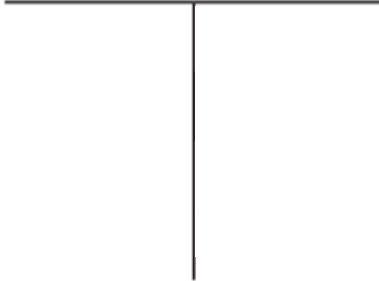
Raw
Materials



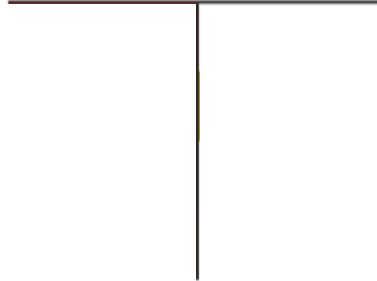
Work in Process
Melting Dept.



Wages
Payable



Work in Process
Skim/Alloy Dept.



Finished
Goods



Factory
Overhead



Work in Process
Mold/Extrude Dept.



Steps and cost and production information needed for each department/ process to prepare the production cost report		Examples from chapter 5
1	Total costs to account for during the period: • Materials • Conversion costs	EA12
2	The number of product units at different stages of the production process: • Beginning WIP inventory • Started during the period • Completed and transferred out • Ending WIP inventory	EA1, EA7, and EA8
3	The number of equivalent units of production for: • Completed and transferred: materials and conversion • Ending WIP inventory: materials and conversion	EA3, EA4, EA9, and EA10
4	Cost per equivalent unit: • Materials • Conversion	PA2 and EA6
5	Allocate the costs determine in step 1 to: a) Units completed and transferred out, and b) Partially completed units in the work in process inventory	EA5, EA12, EA14, EA15, and PB10



**Determine the Total Costs to
Account for During the Period**

Chapter 5. EA 12. What are the total costs to account for if a company's beginning inventory had \$231,432 in materials, \$186,450 in conversion costs, and added direct material costs (\$4,231,392), direct labor (\$2,313,392), and manufacturing overhead (\$1,156,696)?

Solution



Determine the # of Units at Different Stages of Manufacturing Process

Chapter 5. EA 1. How many units were started into production in a period if there were zero units of beginning work in process inventory, 1,100 units in ending work in process inventory, and 21,500 completed and transferred out units?

Solution

Chapter 5. EA7. How many units must be in ending inventory if beginning inventory was 15,000 units, 55,000 units were started, and 57,000 units were completed and transferred out?

Chapter 5. EA8. How many units must have been completed and transferred if beginning inventory was 75,000 units, ending inventory was 72,000 units, and 290,000 units were started?

Solution

	EA7	EA8



Equivalent Units

Equivalent Units (Weighted Average)

- Essentially, the concept of equivalent units involves expressing a given number of partially completed units as a _____ number of fully completed units.
- We do this because it is easier to account for whole units than parts of a unit. We are adding together partially completed units to make a whole unit.
 - For example, if we have ____ units _____ of the way complete, we can add them together to make ____ equivalent unit (= _____).
 - We can make this calculation easier by multiplying the units by a _____ of complete.

Equivalent Units (Weighted Average)

- Under the weighted average method, equivalent units are calculated based on two things:
 1. **Units completed and transferred out, and**
 2. **Units in ending work in process inventory.**
- Units completed and transferred out will always be _____% complete for equivalent unit calculations for direct materials, direct labor, and overhead.
- For units in ending work in process, we would take the units unfinished x a percent complete. The percent complete can be _____ for direct materials, direct labor, or overhead.

Determine the # of Equivalents Units of Production for Materials and Conversion

Chapter 5. EA 3. Given the following information, determine the equivalent units of ending WIP for materials and conversion under the weighted-average method:

- beginning inventory of 2,500 units is 100% complete with regard to materials and 60% complete with regard to conversion
- 18,000 units were started during the period
- 17,500 units were completed and transferred
- ending inventory is 100% complete with materials and 65% complete with conversion

Solution

Chapter 5. EA 4. There were 1,700 units in beginning inventory that were 40% complete with regard to conversion. During the month, 8,550 units were started and 9,000 were transferred to finished goods. The ending work in process was 60% complete with regard to conversion costs, and materials are added at the beginning of the process. What is the total amount of equivalent units for materials and conversion at the end of the month using the weighted-average method?

Solution

Chapter 5. EA 9. Using the weighted-average method, compute the equivalent units of production if the beginning inventory consisted of 20,000 units; 55,000 units were started in production; and 57,000 units were completed and transferred to finished goods inventory. For this process, materials are added at the beginning of the process, and the units are 35% complete with respect to conversion.

Solution

Chapter 5. EA 10. Using the weighted-average method, compute the equivalent units of production for a new company that started 85,000 units into production and transferred 67,000 to the second department. Assume that beginning inventory was 0. Conversion is considered to occur evenly throughout the process, while materials are added at the beginning of the process. The ending inventory for Equivalent Units: Conversion is 9,000 units.

Solution



Determine the Cost per Equivalent Unit of Production

Solution

24

Chapter 5. EA 6. There were 2,400 units in ending work in process inventory that were 100% complete with regard to material and 25% complete with regard to conversion costs. Ending work in process inventory had a cost of \$9,000 and a per-unit material cost of \$2. What was the conversion cost per unit using the weighted-average method?

Solution

**Determine the Cost or Value
of Units Completed and
Transferred out and
Ending WIP Inventory**

Chapter 5. EA 2. A company started a new product, and in the first month started 100,000 units. The ending work in process inventory was 20,000 units that were 100% complete with materials and 75% complete with conversion costs. There were 100,000 units to account for, and the equivalent units for materials was \$6 per unit while the equivalent units for conversion was \$8 per unit. What is the value of the inventory transferred out, using the weighted-average inventory method?

Solution

Chapter 5. EA 5. A company has 1,500 units in ending work in process that are 30% complete after transferring out 10,000 units. All materials are added at the beginning of the process. If the cost per unit is \$4 for materials and \$7 for conversion, what is the cost of units transferred out and in ending work in process inventory using the weighted-average method?

Solution

Chapter 5. EA 14. A production department within a company received materials of \$10,000 and conversion costs of \$10,000 from the prior department. It added material of \$27,200 and conversion costs of \$53,000. The equivalent units are 20,000 for material and 18,000 for conversion. What is the unit cost for materials and conversion?

Solution

Chapter 5. EA 15. Production data show 35,920 units were transferred out of a stage of production and 6,150 units remained in ending WIP inventory that was 100% complete to material and 35% complete to conversion. The unit material cost is \$5 for material and \$8 for conversion. What is the amount of inventory transferred out and remaining in ending work in process inventory?

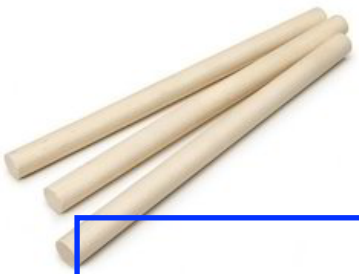
Solution

Chapter 5. PB10. The packaging department began the month with 750 units that were 100% complete with regard to material and 25% complete with regard to conversion. It received 9,500 units from the processing department and ended the month with 500 units that were 100% complete with regard to materials and 75% complete with regard to conversion. With a \$7 per unit material cost and a \$4 per unit cost for conversion, what is the cost of the units transferred out and remaining in ending inventory?

Solution



Prepare Production Cost Report



Rock City Percussion - Manufacturing Process

Shaping Department

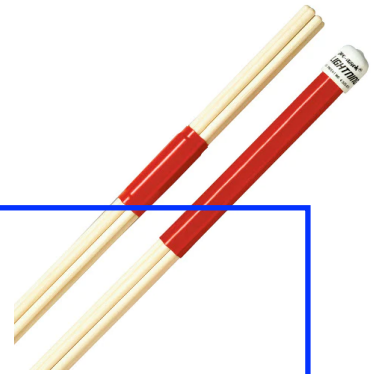
- Direct material: wood
- Indirect material: water
- Wood => *cut* underwater => dowels => *separate* => *shape* => drumsticks
- Machines (electricity); personnel (wage)



Conveyer belt (one batch at a time)
(drumsticks and their related costs)

Finishing/Packaging Department

- Direct material: packaging sleeves (basic size 5A stick); nylon, felt, and/or the ingredients (other types)
- DL and MOH: testing, weighing, and sound-matching the drumsticks into pairs; packaging



EXAMPLE 1. During the month, Rock City Percussion's shaping department requested \$10,179 in direct material and started into production 8,700 hickory drumsticks of size 5A. There was no beginning inventory in the shaping department, and 7,500 drumsticks were completed in that department and transferred to the finishing department. Wood is the only direct material in the shaping department, and it is added at the beginning of the process, so the work in process (WIP) is considered to be 100% complete with respect to direct materials. At the end of the month, the drumsticks still in the shaping department were estimated to be 35% complete with respect to conversion costs. All materials are added at the beginning of the shaping process.

While beginning the size 5A drumsticks, the shaping department incurred these costs in July:

Direct materials \$10,179; direct labor \$15,176; applied overhead \$7,000; total costs \$32,355 (= DM + DL + MOH).

Prepare a **Production Cost Report** for the shaping department using weighted-average method.

Production Cost Report - Shaping Department

Production Cost Report:	Units	DM (\$, % complete, or EU)	CC (\$, % complete, or EU)
1. Total costs incurred:			
Beginning WIP inventory (units and \$)			
Started into production this year (units and \$)			
2. Total equivalent units produced from those costs:			
Completed and transferred out (% complete)			
Completed and transferred out (# of EU)			
Ending WIP inventory (% complete)			
Ending WIP inventory (# of EU)			
Total equivalent units			
3. Costs per equivalent unit			
4a. Value of units completed and transferred out			
4b. Value of ending WIP inventory			

Production Cost Report for the Shaping Department

Units to account for	Units		
Beginning work in process			
Units started into production			
Total units to account for			
Work in process completion %			
Units accounted for	Materials Units	Conversion Units	Total
Completed and transferred out			
Ending work in process			
Total units to account for			
Costs to account for	Materials	Conversion	Total
Beginning work in process			
Incurred during the period			
Total costs to account for			
Equivalent units			
Cost per equivalent unit			
	Materials	Conversion	Total
Value of ending work in process			
Completed and transferred			
Total costs			

Production Cost Report for the Shaping Department

Units to account for	Units
Beginning work in process	—
Units started into production	8,700
Total units to account for	<u>8,700</u>
Work in process completion %	100% 35%

Units accounted for	Materials Units	Conversion Units	Total
Completed and transferred out	7,500	7,500	7,500
Ending work in process	<u>1,200</u>	<u>420</u>	<u>1,200</u>
Total units to account for	<u>8,700</u>	<u>7,920</u>	<u>8,700</u>

Costs to account for	Materials	Conversion	Total
Beginning work in process	\$ 0	\$ 0	\$ 0
Incurred during the period	<u>\$10,179</u>	<u>\$22,176</u>	<u>\$32,355</u>
Total costs to account for	<u>\$10,179</u>	<u>\$22,176</u>	<u>\$32,355</u>
Equivalent units	8,700	7,920	
Cost per equivalent unit	\$ 1.17	\$ 2.80	\$ 3.97

	Materials	Conversion	Total
Value of ending work in process	\$ 1,404	\$ 1,176	\$ 2,580
Completed and transferred	<u>8,775</u>	<u>21,000</u>	<u>29,775</u>
Total costs	<u><u>\$10,179</u></u>	<u><u>\$22,176</u></u>	<u><u>\$32,355</u></u>

Chapter 5, PA 5

Materials are added at the beginning of a production process, and ending work in process inventory is 30% complete with respect to conversion costs. Use the information provided to complete a production cost report using the weighted-average method.

Costs to Account For

Beginning inventory: materials	\$ 10,000
Beginning inventory: conversion	19,000
Direct materials	50,000
Direct labor	75,000
Applied overhead	<u>37,248</u>
Total costs to account for	\$191,248

Units to Account For

Beginning work in process	5,000
Units started into production	20,000
Transferred out	19,000

Production Cost Report

Production Cost Report:	Units	DM (\$, % complete, or EU)	CC (\$, % complete, or EU)
1. Total costs incurred:			
Beginning WIP inventory (units and \$)			
Started into production this year (units and \$)			
2. Total equivalent units produced from those costs:			
Completed and transferred out (% complete)			
Completed and transferred out (# of EU)			
Ending WIP inventory (% complete)			
Ending WIP inventory (# of EU)			
Total equivalent units			
3. Costs per equivalent unit			
4a. Value of units completed and transferred out			
4b. Value of ending WIP inventory			

Chapter 5, PA 3

Pant Risers manufactures bands for self-dressing assistive devices for mobility-impaired individuals. Manufacturing is a one-step process where the bands are cut and sewn. This is the information related to this year's production:

Units to Account For	Units	Materials	Conversion
Beginning WIP inventory	500	500	250
Started	<u>20,500</u>		
To account for	21,000		

Ending inventory was 100% complete as to materials and 70% complete as to conversion, and the total materials cost is \$57,540 and the total conversion cost is \$36,036. Using the weighted-average method, what are the unit costs if the company transferred out 17,000 units? What is the value of the inventory transferred out and the value of the ending WIP inventory?

Production Cost Report

Production Cost Report:	Units	DM (\$, % complete, or EU)	CC (\$, % complete, or EU)
1. Total costs incurred:			
Beginning WIP inventory (units and \$)			
Started into production this year (units and \$)			
2. Total equivalent units produced from those costs:			
Completed and transferred out (% complete)			
Completed and transferred out (# of EU)			
Ending WIP inventory (% complete)			
Ending WIP inventory (# of EU)			
Total equivalent units			
3. Costs per equivalent unit			
4a. Value of units completed and transferred out			
4b. Value of ending WIP inventory			



Journal Entries for a Process Costing System

Chapter 5. EA 16. Overhead is assigned to the manufacturing department at the rate of \$10 per machine hour. There were 3,500 machine hours during October in the shaping department and 2,500 in the packaging department. Prepare the journal entry to apply overhead to the manufacturing departments.

Chapter 5. EA 17. Prepare the journal entry to record the factory wages of \$28,000 incurred for a single production department assuming payment will be made in the next pay period.

Solution

Chapter 5. EA 18. Prepare the journal entry to record the transfer of 3,000 units from the packaging department to finished goods if the material cost per unit is \$4 and the conversion cost per unit is \$5.50.

Chapter 5. EA 19. Prepare the journal entry to record the sale of 2,000 units that cost \$8 per unit and sold for \$15 per unit.

Solution

[illegible]

Chapter 5, PB 13

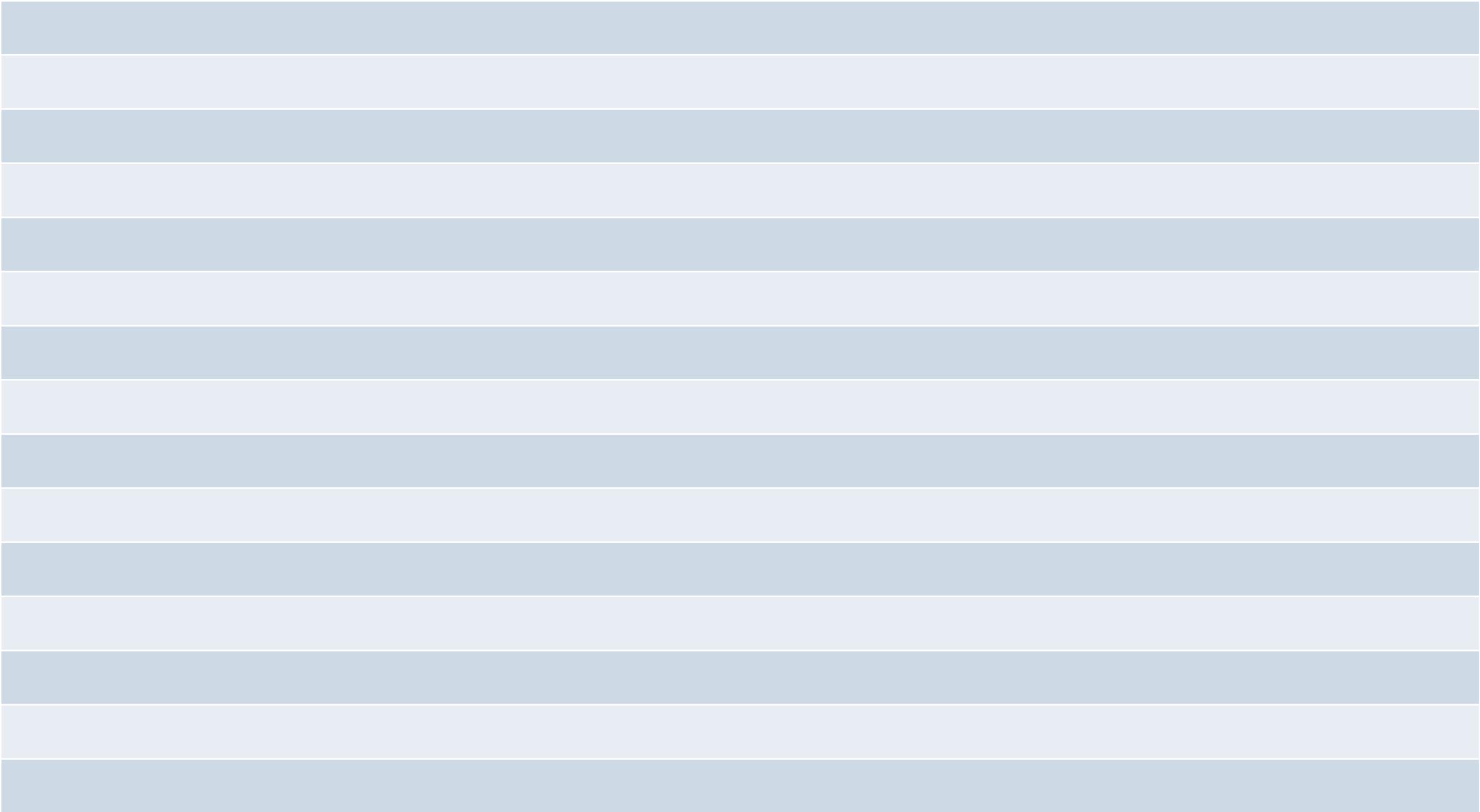
Selected information from Hernandez Corporation shows the following:

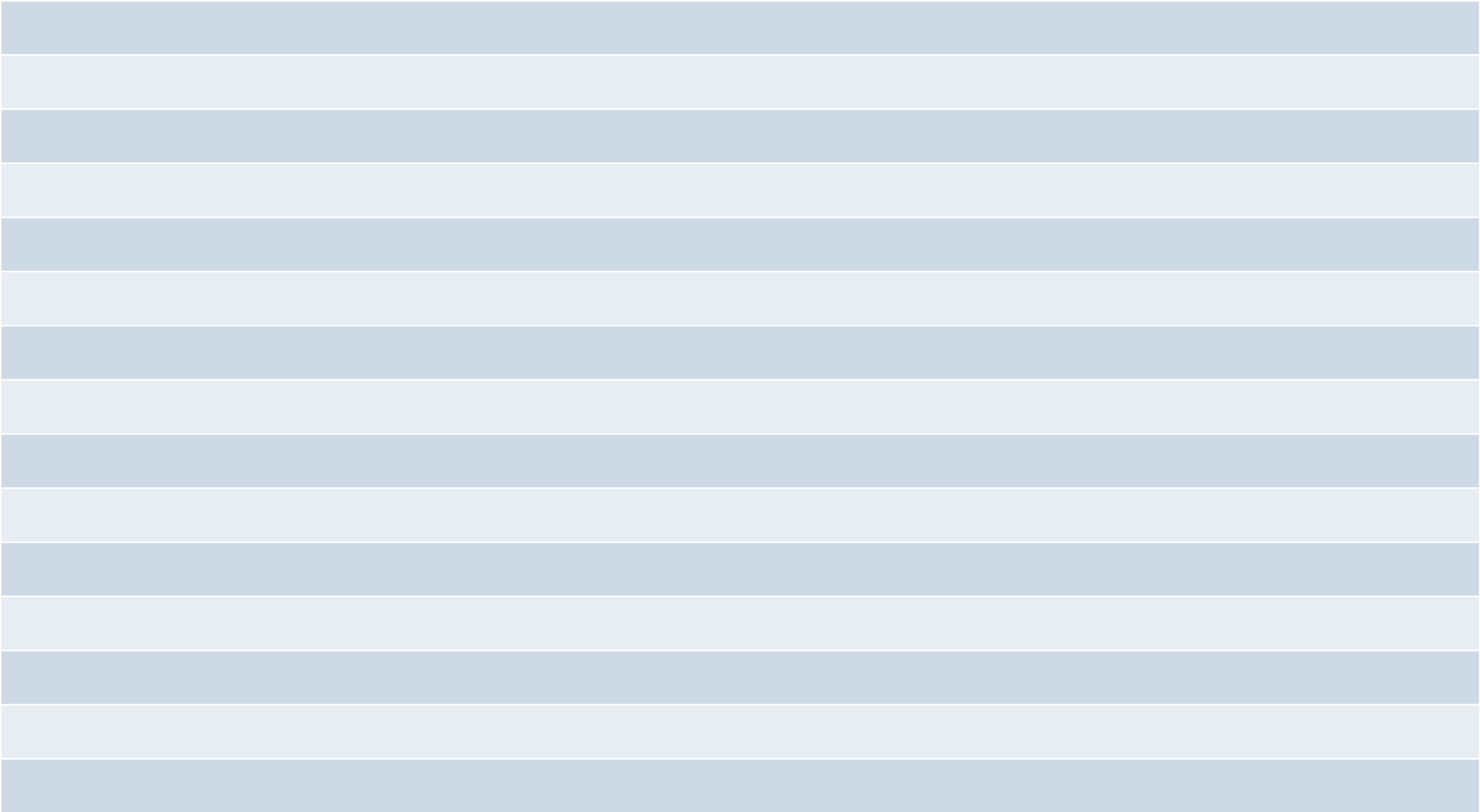
Raw materials purchased	\$12,000
Direct materials	\$ 3,500
Direct labor	\$14,000
Factory depreciation	\$ 4,200
Total manufacturing overhead incurred	\$18,500
Machine hours per month	34,000
Unit cost for material	\$ 2.75
Unit cost of conversion	\$ 5.95
Number of units transferred	16,500

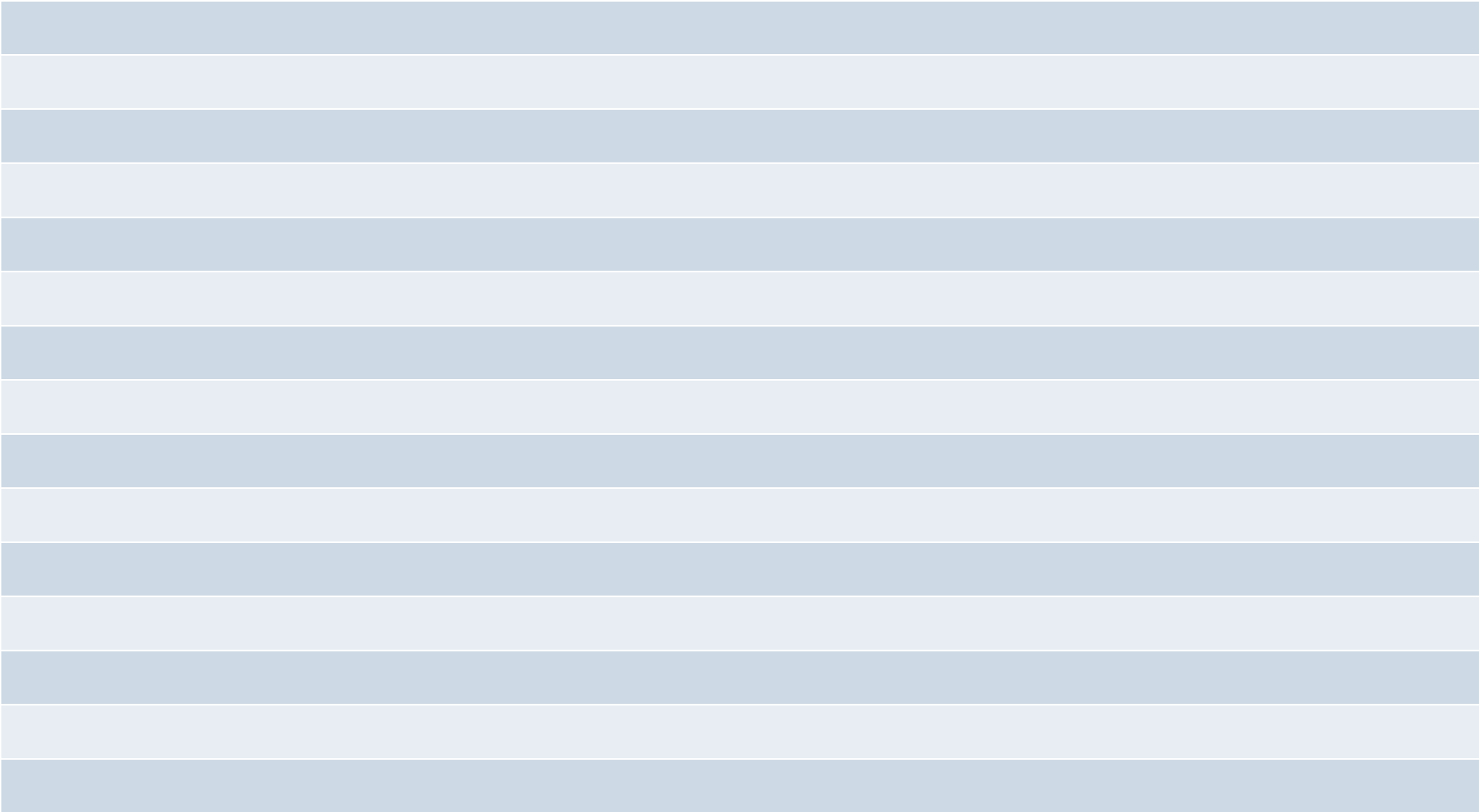
Prepare journal entries to record the following:

- A. raw material purchased
- B. direct labor incurred
- C. depreciation expense (hint: this is part of manufacturing overhead)
- D. raw materials used
- E. overhead applied on the basis of \$0.50 per machine hour
- F. the transfer from department 1 to department 2

Description	DR	CR







Chapter 5, PB 13

Selected information from Hernandez Corporation shows the following:

Raw materials purchased	\$12,000
Direct materials	\$ 3,500
Direct labor	\$14,000
Factory depreciation	\$ 4,200
Total manufacturing overhead incurred	\$18,500
Machine hours per month	34,000
Unit cost for material	\$ 2.75
Unit cost of conversion	\$ 5.95
Number of units transferred	16,500

Prepare journal entries to record the following:

- A. raw material purchased
- B. direct labor incurred
- C. depreciation expense (hint: this is part of manufacturing overhead)
- D. raw materials used
- E. overhead applied on the basis of \$0.50 per machine hour
- F. the transfer from department 1 to department 2

Description	DR	CR

Process Costing: Summary

- The optimal system for a company to use when the production process results in many similar units.
- Used when production is continuous or occurs in large batches and it is difficult to trace a particular input cost to a specific individual product.
- Output: Production Cost Report
- In the report, the following information is reported:
 - Cost per equivalent unit
 - Costs to units completed and transferred out of the department or process
 - Costs to partially completed units in the work in process inventory

REVIEW

- The three components of manufacturing costs are _____, _____, and _____.
- Job order costing is the optimal costing method for producing _____ goods or when it is easy to identify the cost _____ with the product. MOH is assigned to the individual jobs using _____.
- Process costing is the system of accumulating costs within each _____ for large-volume, mass-produced units. Process costing determines the cost per unit through the use of _____.
- Selling and administrative costs: treated as _____ costs in traditional costing systems; related to the sales of products and management of the company and are _____ directly tied to a specific product.
- Traditional methods is optimal when the manufacturing process is _____ driven and overhead increases based on traditional activity bases, such as direct labor hours, direct labor dollars, or machine hours.



Thank you!
Questions?